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Question 1

Capture and analyze ARP packets using Wireshark. Inspect the ARP request and reply frames when your device attempts to find the router's MAC address.

Discuss the importance of ARP in packet forwarding

I have opened my wireshark and selected the WiFi(Hostel) to inspect the ARP Request and the ARP reply frames by entering 'arp' in the filter bar.

I was pinging another PC with the ip address 192.168.220.1

7939	483.257671	AzureWaveTec_25:72:…	be:79:10:1c:24:a8	ARP	42	172.20.24.62 is at 14:13:33:25:72:17
7942	483.973797	be:79:10:1c:24:a8	AzureWaveTec_25:72:…	ARP	56	Who has 172.20.24.62? Tell 172.20.24.67
7943	483.973818	AzureWaveTec_25:72:…	be:79:10:1c:24:a8	ARP	42	172.20.24.62 is at 14:13:33:25:72:17
7954	484.997560	be:79:10:1c:24:a8	AzureWaveTec_25:72:…	ARP	56	Who has 172.20.24.62? Tell 172.20.24.67
7955	484.997627	AzureWaveTec_25:72:…	be:79:10:1c:24:a8	ARP	42	172.20.24.62 is at 14:13:33:25:72:17
7969	487.249678	be:79:10:1c:24:a8	AzureWaveTec_25:72:…	ARP	56	Who has 172.20.24.62? Tell 172.20.24.67
7970	487.249724	AzureWaveTec_25:72:…	be:79:10:1c:24:a8	ARP	42	172.20.24.62 is at 14:13:33:25:72:17
7971	488.069987	be:79:10:1c:24:a8	AzureWaveTec_25:72:…	ARP	56	Who has 172.20.24.62? Tell 172.20.24.67
7972	488.070037	AzureWaveTec_25:72:…	be:79:10:1c:24:a8	ARP	42	172.20.24.62 is at 14:13:33:25:72:17
7975	488.989950	be:79:10:1c:24:a8	AzureWaveTec_25:72:…	ARP	56	Who has 172.20.24.62? Tell 172.20.24.67
7976	488.989967	AzureWaveTec_25:72:…	be:79:10:1c:24:a8	ARP	42	172.20.24.62 is at 14:13:33:25:72:17
8108	495.243711	be:79:10:1c:24:a8	AzureWaveTec_25:72:…	ARP	56	Who has 172.20.24.62? Tell 172.20.24.67
8109	495.243765	AzureWaveTec_25:72:…	be:79:10:1c:24:a8	ARP	42	172.20.24.62 is at 14:13:33:25:72:17
8110	495.957192	be:79:10:1c:24:a8	AzureWaveTec_25:72:…	ARP	56	Who has 172.20.24.62? Tell 172.20.24.67
8111	495.957240	AzureWaveTec_25:72:…	be:79:10:1c:24:a8	ARP	42	172.20.24.62 is at 14:13:33:25:72:17
8112	496.971058	be:79:10:1c:24:a8	AzureWaveTec_25:72:…	ARP	56	Who has 172.20.24.62? Tell 172.20.24.67
8113	496.971107	AzureWaveTec_25:72:…	be:79:10:1c:24:a8	ARP	42	172.20.24.62 is at 14:13:33:25:72:17
12473	801.003067	be:79:10:1c:24:a8	AzureWaveTec_25:72:…	ARP	56	Who has 172.20.24.62? Tell 172.20.24.67

The Wireshark capture shows two types of ARP packets:

ARP Request

Example captured packet:

Who has 172.20.24.62? Tell 172.20.24.67

This indicates that the device with IP address **172.20.24.67** is requesting the MAC address of the device with IP address **172.20.24.62**.

This request is **broadcast to all devices** in the local network.

Field	Value
Sender IP Address	172.20.24.67
Target IP Address	172.20.24.62
Sender MAC Address	be:79:10:1c:24:a8
Destination MAC	Broadcast

ARP Reply

Example captured packet:

172.20.24.62 is at 14:13:33:25:72:17

This indicates that the device **172.20.24.62** is replying with its MAC address.

Important details:

Field	Value
Sender IP Address	172.20.24.62
Sender MAC Address	14:13:33:25:72:17
Target IP Address	172.20.24.67

The ARP reply is sent **directly to the requesting device**.

Analysis

When a device wants to send a packet to another device within the same network, it must know the **MAC address** of the destination device. However, communication is usually initiated using **IP addresses**. ARP is used to translate the IP address into a MAC address.

The process works as follows:

1. The sender checks its **ARP cache** to see if the MAC address is already known.
2. If the MAC address is not available, the sender broadcasts an **ARP request** to the network.
3. The device that owns the requested IP address sends an **ARP reply** with its MAC address.
4. The sender stores this information in the **ARP table** for future communication.

Importance of ARP in Packet Forwarding

ARP plays a crucial role in packet forwarding because network communication at the **data link layer requires MAC addresses**, while applications use **IP addresses**. ARP acts as a bridge between these two layers by mapping IP addresses to MAC addresses.

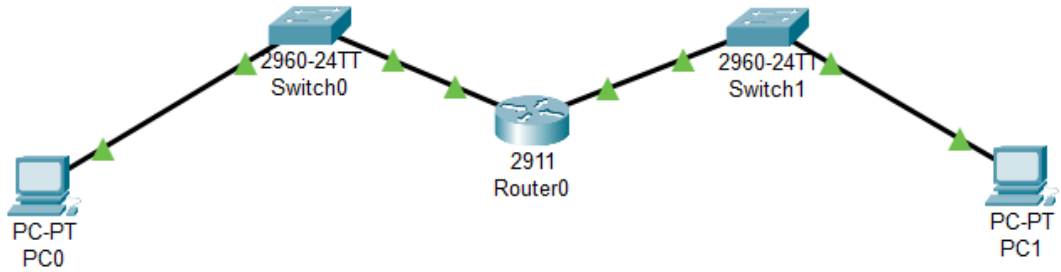
Without ARP, devices in a local network would not be able to determine the correct hardware address required to deliver Ethernet frames, making communication impossible.

Question 2

Manually configure static routes on a router to direct packets to different subnets.

Use the ip route command and verify connectivity using ping and traceroute.

<u>The simple layout in Cisco Packet Tracer:</u>

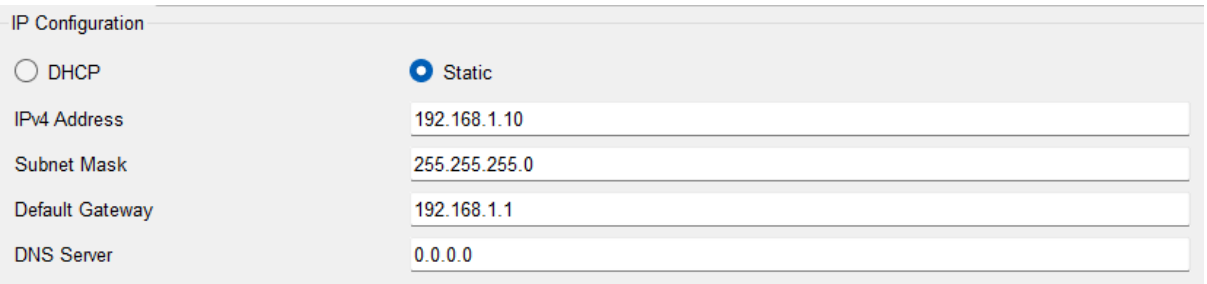


Configuring the router:

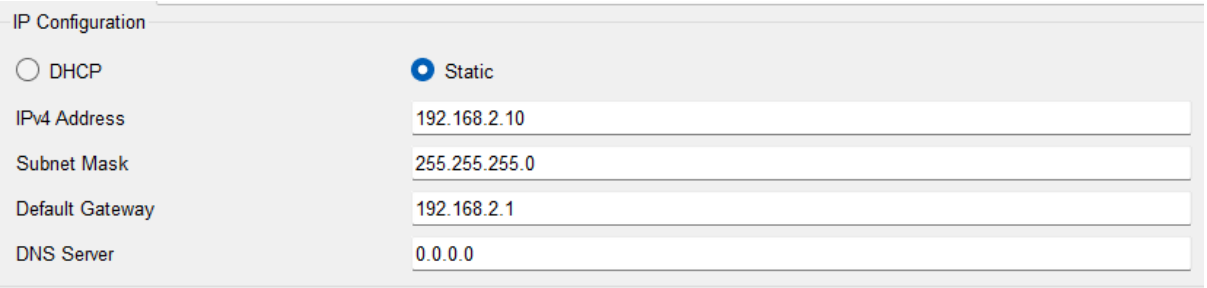
```
Router>enable
Router#config terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown

Router(config)#interface g0/1
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
exit
```

Configuring the PCs:

PC0:



PC1:



Therefore, the static routing is done. Now checking the connectivity.

Pinging PC1 from PC0:

```
C:\>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Reply from 192.168.2.10: bytes=32 time<1ms TTL=127
Reply from 192.168.2.10: bytes=32 time<1ms TTL=127
Reply from 192.168.2.10: bytes=32 time<1ms TTL=127
Reply from 192.168.2.10: bytes=32 time=1ms TTL=127

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Pinging PC0 from PC1:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.10

Pinging 192.168.1.10 with 32 bytes of data:

Reply from 192.168.1.10: bytes=32 time<1ms TTL=127
Reply from 192.168.1.10: bytes=32 time<1ms TTL=127
Reply from 192.168.1.10: bytes=32 time<1ms TTL=127
Reply from 192.168.1.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

The router interfaces were configured with IP addresses for two different subnets. Since both networks were directly connected to the router, the router automatically added these networks to its routing table as connected routes. Therefore, static routing was not required in this topology. Connectivity between the two networks was successfully verified using the ping command.

Question 3

Given a network address of 10.0.0.0/24, divide it into 4 equal subnets.

- **Calculate the new subnet mask**
- **Determine the valid host range for each subnet**
- **Assign IP addresses to devices in Packet Tracer**
- **Verify connectivity**

3. The given network address: 10.0.0.0/24

Total no of address : $2^{32-24} = 2^8 = 256$

We need to divide that by 4:

$$\therefore \frac{256}{4} = 64 = 2^6$$

$\therefore$ New subnet mask = $32 - x = 6$

$x = 26$

$\therefore$ New subnet mask : 10.0.0.0/26

$\therefore$ New subnet task : 255.255.255.192

Subnet	Network Address	Host Range	Broadcast Address
Subnet 1	10.0.0.0/26	10.0.0.1 to 10.0.0.62	10.0.0.63
Subnet 2	10.0.0.64/26	10.0.0.65 to 10.0.0.126	10.0.0.127
Subnet 3	10.0.0.128/26	10.0.0.129 to 10.0.0.190	10.0.0.191
Subnet 4	10.0.0.192/26	10.0.0.193 to 10.0.0.254	10.0.0.255

Device	IP Address	Subnet Mask	Default Gateway
PC0	10.0.0.2	255.255.255.192	10.0.0.1
PC1	10.0.0.66	255.255.255.192	10.0.0.65
PC2	10.0.0.130	255.255.255.192	10.0.0.129
PC3	10.0.0.194	255.255.255.192	10.0.0.193

The network 10.0.0.0/24 was successfully divided into 4 equal subnets using the /26 subnet mask. Each subnet contains 62 valid host addresses. IP addresses were assigned from each subnet, and connectivity can be verified by testing communication between devices.

The connection checking:

Pinging from PC4 to PC6:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.0.0.2

Pinging 10.0.0.2 with 32 bytes of data:

Reply from 10.0.0.2: bytes=32 time=1ms TTL=128
Reply from 10.0.0.2: bytes=32 time=9ms TTL=128
Reply from 10.0.0.2: bytes=32 time=6ms TTL=128
Reply from 10.0.0.2: bytes=32 time=8ms TTL=128

Ping statistics for 10.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 9ms, Average = 6ms
```

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.0.0.66

Pinging 10.0.0.66 with 32 bytes of data:

Reply from 10.0.0.66: bytes=32 time=7ms TTL=128
Reply from 10.0.0.66: bytes=32 time=4ms TTL=128
Reply from 10.0.0.66: bytes=32 time=3ms TTL=128
Reply from 10.0.0.66: bytes=32 time=12ms TTL=128

Ping statistics for 10.0.0.66:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 3ms, Maximum = 12ms, Average = 6ms
```

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.0.0.130

Pinging 10.0.0.130 with 32 bytes of data:

Reply from 10.0.0.130: bytes=32 time=7ms TTL=128
Reply from 10.0.0.130: bytes=32 time<1ms TTL=128
Reply from 10.0.0.130: bytes=32 time=1ms TTL=128
Reply from 10.0.0.130: bytes=32 time<1ms TTL=128

Ping statistics for 10.0.0.130:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 7ms, Average = 2ms
```

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.0.0.194

Pinging 10.0.0.194 with 32 bytes of data:

Reply from 10.0.0.194: bytes=32 time=5ms TTL=128
Reply from 10.0.0.194: bytes=32 time<1ms TTL=128
Reply from 10.0.0.194: bytes=32 time=6ms TTL=128
Reply from 10.0.0.194: bytes=32 time=17ms TTL=128

Ping statistics for 10.0.0.194:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 17ms, Average = 7ms
```

By this way, the pinging will ensure the perfect connectivity.

Question 4

You are given three IP addresses: 192.168.10.5, 172.20.15.1, and 8.8.8.8.

- **Identify the class of each IP address.**
- **Determine if it is private or public.**
- **Explain how NAT would handle a private IP when accessing the internet.**

IP Address Classification

1. IP Address: 192.168.10.5

- **Class:** Class C
- **Reason:** Class C addresses range from **192.0.0.0 to 223.255.255.255**
- **Type:** Private IP
- **Reason:** It belongs to the private IP range **192.168.0.0 to 192.168.255.255**

2. IP Address: 172.20.15.1

- **Class:** Class B
- **Reason:** Class B addresses range from **128.0.0.0 to 191.255.255.255**
- **Type:** Private IP
- **Reason:** It belongs to the private IP range **172.16.0.0 to 172.31.255.255**

3. IP Address: 8.8.8.8

- **Class:** Class A
- **Reason:** Class A addresses range from **1.0.0.0 to 126.255.255.255**
- **Type:** Public IP
- **Reason:** It does not belong to any private IP range and is globally routable on the internet

IP Address	Class	Private/Public
192.168.10.5	Class C	Private
172.20.15.1	Class B	Private
8.8.8.8	Class A	Public

How NAT Handles a Private IP

Private IP addresses cannot directly communicate over the internet because they are used only inside local networks. To allow devices with private IP addresses to access the internet, a router uses **NAT (Network Address Translation)**.

For example, if a device with private IP **192.168.10.5** wants to access **8.8.8.8**, the router performs the following steps:

1. The internal device sends a packet with source IP **192.168.10.5** and destination IP **8.8.8.8**
2. The router replaces the private source IP with its own **public IP address**
3. The router stores this mapping in a NAT table
4. The packet is then forwarded to the internet
5. When the reply comes back, the router checks the NAT table and sends the response back to the original private device

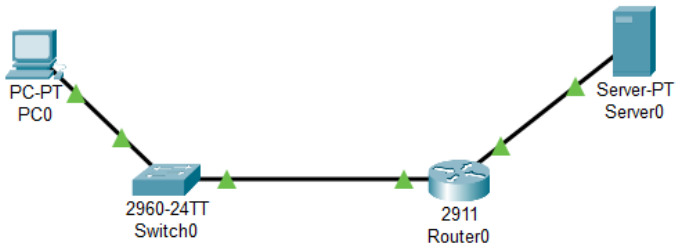
Thus, NAT allows multiple private devices to share one public IP address for internet access.

Question 5

In **Cisco Packet Tracer**, configure **NAT on a router** to allow internal devices (**192.168.1.x**) to access the internet.

- Test connectivity by **pinging an external public IP**
- Capture the traffic in **Wireshark** and analyze the **source IP before and after NAT translation**

The simple topology:



The NAT configuration in router:

I configures both NAT inside and NAT outside:

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#ip nat inside
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
Router(config)#interface g0/1
Router(config-if)#ip address 203.0.113.1 255.255.255.0
Router(config-if)#ip nat outside
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
Router(config)#
```

Configuring the Access Control List(ACL) and the NAT overload:

```
Router(config)#
Router(config)#access-list 1 permit 192.168.1.0 0.0.0.255
Router(config)#ip nat inside source list 1 interface g0/1 overload
Router(config)#
```

I got this output:

Inside global	Inside local	Outside local	Outside global
203.0.113.1	192.168.1.10	203.0.113.2	203.0.113.2

Field	**Meaning**
Inside Local	Private IP (PC)
Inside Global	Router public IP
Outside Global	Destination IP

Therefore, the NAT is working.

We can the NAT statistics:

```
Router#show ip nat statistics
Total translations: 0 (0 static, 0 dynamic, 0 extended)
Outside Interfaces: GigabitEthernet0/1
Inside Interfaces: GigabitEthernet0/0
Hits: 0  Misses: 5
Expired translations: 0
Dynamic mappings:
```

This means:

- NAT is configured, but no packets were translated yet.
- The router did not perform any NAT translation

NAT was successfully configured on the router to allow internal devices in the private network (192.168.1.x) to communicate with external networks. The router translated the private IP address into its public IP address before forwarding packets. Connectivity was verified using the ping command, and NAT translation was confirmed using the show ip nat translations command.